



Data Analysis, Tasks, Personas

Introduction to HCI

Hasti Seifi

Arizona State University

Learning Objectives

- Explain main considerations in data collection
- Describe two types of data and data analysis methods
- Create task descriptions for an interactive system
- Select and prioritize tasks and users
- Develop realistic personas
- Distinguish functional and non-functional requirements for an interactive system

Considerations for Data Collection

UX researchers often need to trade-off conflicting criteria

1. Three key criteria

- **Realism:** how similar the data-collection situation is to real world
- **Precision:** how much you can control for irrelevant aspects
- **Generalizability:** how much does it apply beyond your data collection

2. Identifying participants

- Representative users
- PACT analysis

Considerations for Data Collection - Cont.

3. Triangulation

- Collect more than one type of data, e.g., interviews and questionnaires
- Look at data from more than one perspective

4. Ethics

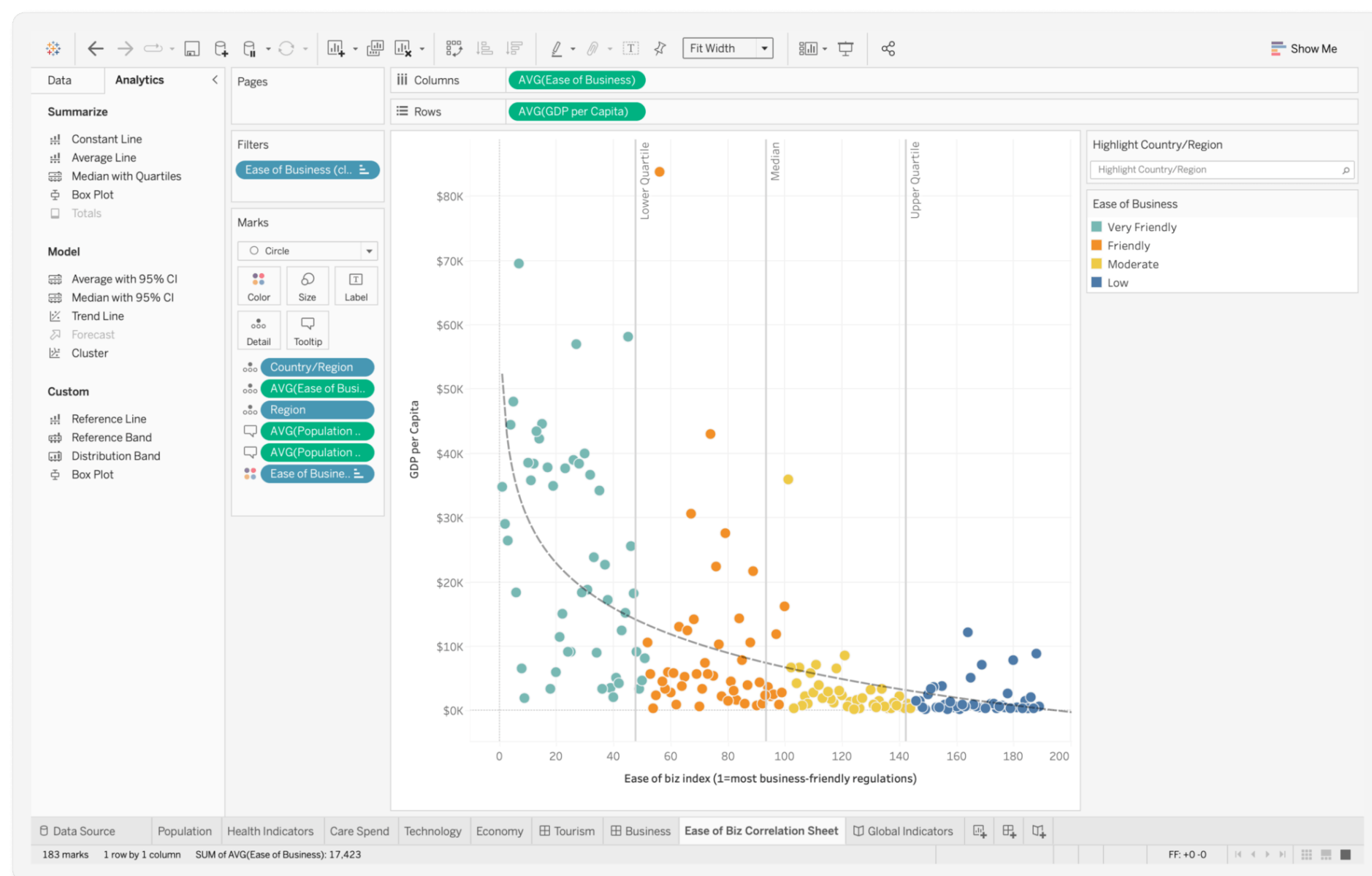
- How you treat participants & their data

Data Analysis

- Data types
 - Quantitative: Numbers
 - Qualitative: Text, images, video
- Types of analysis
 - Quantitative
 - Qualitative
- You can turn qualitative data (e.g., text) into numbers and analyse it with quantitative methods.
 - Example: how many times the user says a positive description for the IT system?
- Important Note: quantitative is not equivalent to objective (or not biased)

Quantitative Data Analysis

- Deals with numbers to find trends
- Plots, charts, interactive visualizations
- Statistical analysis: correlation, significance tests
- Useful for
 - Questionnaire data, controlled experiments
- A variety of software tools available
 - Excel, Tableau, SPSS
 - Programming APIs: R, python, D3.js



Qualitative Data Analysis

- Deals with text, images, video
- Rich description of the patterns in the data
- Frequency of an observation is less important
- A variety of methods
 - Affinity diagramming ← You'll use this
 - Thematic analysis
 - Grounded theory

Qual Analysis: Affinity Diagrams

- Steps for creating affinity diagram
 - a. Record all notes or observations on individual cards or sticky notes
 - b. Look for patterns and notes or observations that are related
 - c. Create a group for each pattern or theme
 - d. Discuss and revise the groups
 - e. Give each theme or group a name

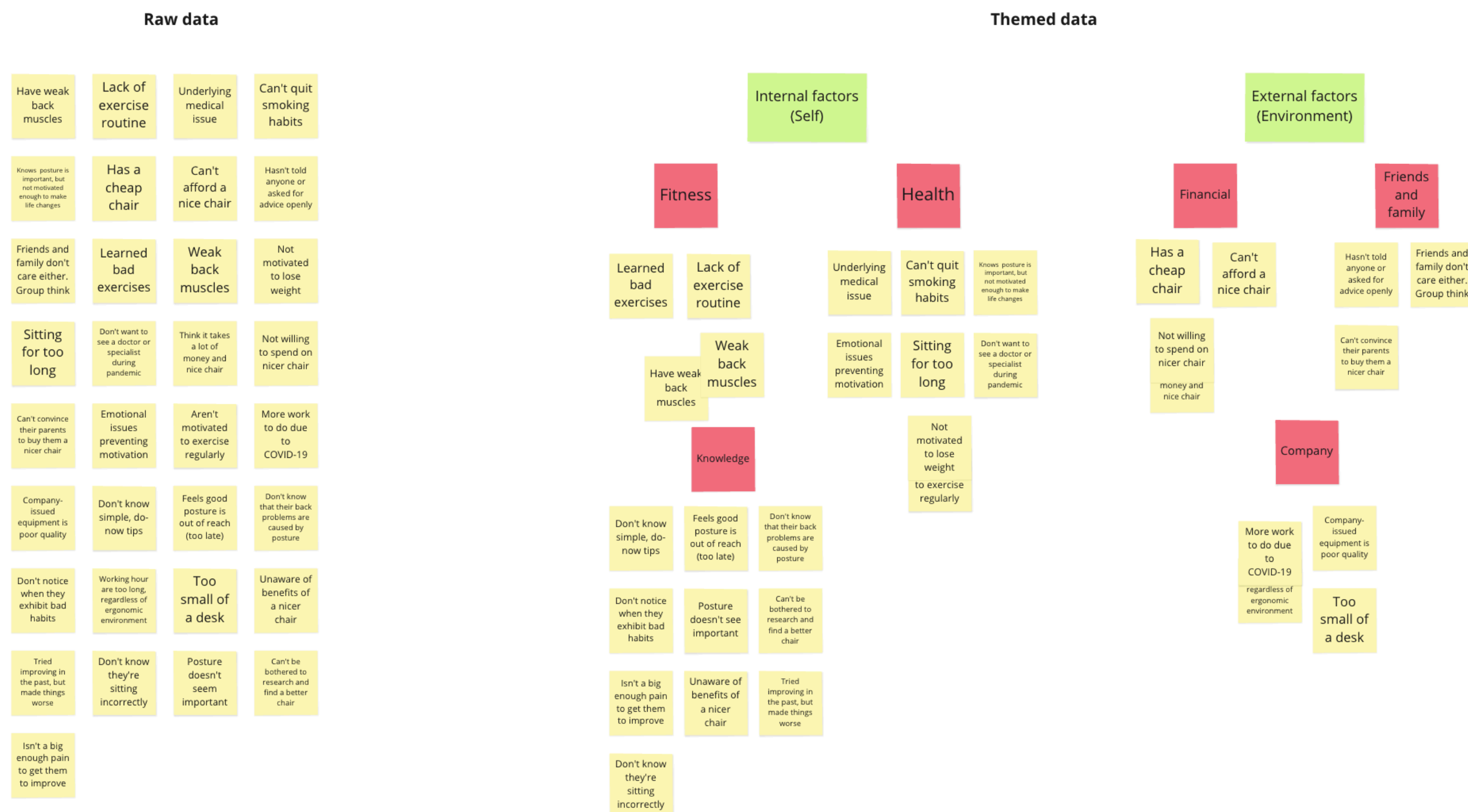


Affinity Diagraming: Miro Board

- Document the process:
 - Combine the activity with immediate writing
 - Relationships between groups
 - One item may relate to two groups

Affinity map example

Problem: Why don't people takes steps to improve their posture?



Qualitative Data Analysis Software

Transcribing Interviews: Otter.ai

The screenshot displays the Otter.ai web application interface. The browser tab is titled 'Eval_P-K03_Non - Otter.ai'. The address bar shows the URL: otter.ai/u/wQ8gCtRIDincqkdHpY2uLoX4ryU?f=%2Fmy-notes%3Fr%3D%252Fmy-notes&tab=summary. The page header includes the Otter.ai logo, the file name 'Eval_P-K03_Non', a search bar, a 'Share' button, and a diamond icon.

The left sidebar contains the following sections:

- User profile: Hasti Seifi, hasti.s.v@gmail.com
- Organization: Arizona State Uni...
- Buttons: '+ Invite teammates'
- Navigation: Home, My Conversations, Shared with Me, More
- GROUPS: General
- DIRECT MESSAGES
- FOLDERS

The main content area displays the transcription summary for 'Eval_P-K03_Non':

- SUMMARY KEYWORDS**: intensity, brush, bit, nodes, explosion, play, sensation, hand, create, scan, feel, save, zigzag pattern, canvas, scale, haptics, primitive, guess, move, loop
- SPEAKERS**: Speaker 1 (54%), Speaker 2 (24%), Speaker 3 (7%)

The transcription list shows three segments:

- Speaker 1** (0:01): Hi. Okay, good, it's working. So, thank you so much for taking the time to participate. Of course, when I ask you some questions about your background, what is your current occupation?
- Speaker 2** (0:17): I'm a **social** worker. I work with adult
- Speaker 1** (0:22): artists. Okay. Can you describe briefly your educational background?

At the bottom, there is a video player interface with a progress bar from 00:18 to 30:15, a play button, a 1x speed control, and icons for transcript, chat, and share.

Qualitative Data Analysis Software

Coding the data: MAXQDA

The screenshot displays the MAXQDA Plus 2018 software interface. The top menu bar includes Home, Import, Codes, Variables, Analysis, Mixed Methods, Visual Tools, Reports, and MAXDictio. Below the menu is a toolbar with icons for New Project, Open Project, Document System, Code System, Document Browser, Retrieved Segments, Logbook, User Management, Save Project As, Save Anonymized Project As, Project from Activated Documents, Merge Projects, Open Exchange File, Export Exchange File, and External Files.

The left sidebar shows the Document System with a list of documents (P1-Transcript to P12-Transcript) and a Code System with a hierarchical structure of codes (Participants, Background, Device Attributes, UX, Engineering, Task/Interaction (user wants)).

The main window displays the Document Browser: P4-Transcript. The text in the document is as follows:

79 So here...(pointing at Haptipedia) so I originally started from both the haply.. here.. (points at Haply)and the Pantograph.. back then which I had already known.. and I kinda tried to connect the dots between.. so the next thing is.. okay.. so Pantograph.. Cool.. uhm.. (thinking).. let's mock it up.. so we mocked it up.. we tried to evaluated first if there's something there.. and then had this been done before.. then we started looking through I need to look at paradigms.. so what I don't see here is the paradigm of interaction.. what that means to me is for dual-panto what I hold-on is this one case... it's this thing where you're exploring environment.. much like you have a stylus and an end effector and you're exploring something that's a force-feedback shape environment.. the other one is a motion-guidance platform- something that leads you around.. people have done like.. done this for Phantom for both of these.. for typical Phantom you explore a 3D shape and you feel it. But sometimes they make sure you had to do tennis racket spin or how to sign a signature for e.g.. so how it was used is something that we need to find out and that this interface can't tell you.. so then next thing I wanna see is like the path.. so I want to see the latest work which is happening know anything in between.. uhm the other thing so my PI had interacted with one.. I think actually fantastic we should get the.. (thinking).. specs called.. but I think Dinesh Pi might have work anymore. So we want to see.. so after I prototype also yeah can I buy it.. commercial product.. s Cool.. so now let's scroll down to find.. oh the peak force vs workspace viz).. (hovers on data missing peak force).. and 1D is.. (scrolls up and peak force go.. (scrolls up and down).. okay for force.. that's Large.. Oh that doesn't tell me.. I Rjo).. (mumbles).. I would like to keep only.. I get back to where I was? (accidentally clicks o

Codes are applied to the text, such as `..Wants to see evolution of device`, `..Quick prototyping/Evaluate fir`, `..How device has been used`, `..Latest Devices as a requirem`, `..Wants to see evolution of de`, `..Commercial Availability`, `..Confused by Peak-force vs Wor`, `..Confused by Peak-force vs Wor`, `..Wants to achieve previous state`, `..Image`, `..Wants to see similar devices`, and `..DoF/DoF Type`.

A "Memo" window is open, showing the document title "Insight", author "ashutosh", last change "2018-06-2", and type "Insight". The memo text reads: "This is similar to what was observed with P2. OS uses those filters/device attributes such that he can explore maximum number of devices within that criteria."

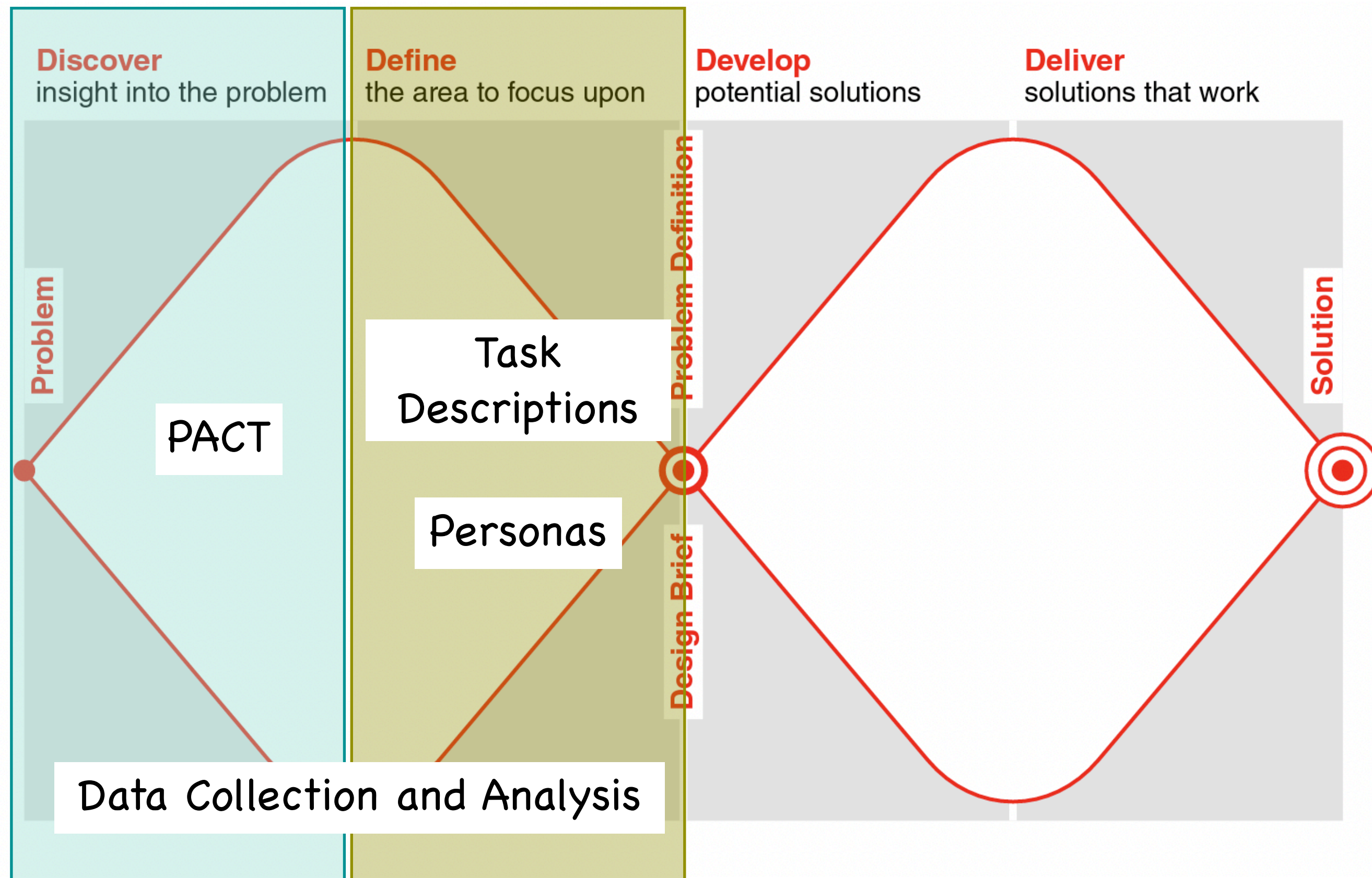
The bottom status bar shows the Simple Coding Query (OR combination of codes).

A4: Data Collection Tips

- You should do interview or interview+X
- Start planning the data collection
 - Who to include?
 - Which method?
 - How to record?
 - Justify your choices
- PACT analysis should guide your data collection
- Pilot your data collection in your team

Now you understand the users and their needs.
What's next?

Human-Centred Design Process



Steps for developing task descriptions

Four steps to identify, validate, and select tasks

1. Discovering the tasks that users do
2. Developing good task descriptions
3. Validating the tasks
4. Deciding which user types and which tasks to include

1. Discovering the Tasks that Users Do

- **The Best:** Observing and/or interviewing the real end user
- **Second best:** Interviewing the end-user representative
- **When all else fails:** Make your beliefs of the end-users and task space explicit.

2. Developing good task descriptions

1. It describes **what the user wants to do** but does **not say how** the user would do it.

- Correct: Emily enters her name and address
- Incorrect: Emily types her name in the textbox on her phone

2. It is **very specific**.

- Correct: Alex browses the catalog and chooses the JPG stroller (cost \$98)
- Incorrect: Alex chooses an item.

2. Developing good task descriptions

3. It describes **a complete job** from when the user starts until they achieve their goal.
4. It says **who the users are** and reflects their real interests.
 - Correct: Fred is a first-time customer to this store and has little computer experience.
5. As a set, the task descriptions identify **a broad coverage of users and their activities.**

Good task description example

Task 1. Fred Johnson, who is caring for his demanding toddler son, wants a good quality umbrella stroller (red is preferred, but blue is acceptable). He browses the catalog and chooses the JPG stroller (cost \$98., item code 323 066 697). He pays for it in cash, and uses it immediately. Fred is a first-time customer to this store, has little computer experience, and says he types very slowly with one finger.

Aaron, a young adult living on his own, is planning simple dinners for the upcoming week. He typically cooks the same easy meals: tacos, quesadillas, and pastas, to keep preparation quick and low effort. He realizes he is running low on vegetables and meats, and since smaller ingredients run out at different times, he often forgets what he already has at home, causing them to expire. After finding the meal planning platform, he scans his fridge and pantry to see what ingredients he has, then notes his ingredients into the app. He selects the recipes for tacos, quesadillas, and pastas. Now, when he goes grocery shopping, he'll know what he already has for these recipes, and won't waste any ingredients buying duplicate ingredients.

Other Bad Examples

Section 2: Tasks

1. **Search for team member reviews left by other users:** Find reviews left by other students filtered by things like class, project, or date.
2. **Read team member reviews left by other users:** Review feedback on a teammate's performance, personality, and work habits.

Task 2: Adding a Household Chore

- **Context:** A student (P2) wants to remind roommates to take out the trash, he also wants the task to be recurring or rotating.
- **Motivation:** Prevent trash buildup and avoid conflict about whose turn it is.
- **Steps:**
 1. Open the app on phone.
 2. Select "Create Task."
 3. Enter "Take out the trash."
 4. Assign to roommate John.
 5. Assign who will do the task next each week i.e. set the task as rotating
 6. Set the task to be recurring
 7. Set deadline to "Tonight, 8 PM."
 8. Submit task, triggering reminder for John.

Task 1. Plan an access aware route. The user needs a shaded route that is quick and free of steps to move between courses. The application starts when the user enters a location. The user selects one of three route cards (Most Shade, Fastest, or No Stairs) which the software applies saved preferences before showing ETAs for approval.

Examples task descriptions

Discuss what aspects of the task descriptions are good or bad?

Task 1: Jessica has listened to her favorite podcast Doug Loves Movies for over two years. She is listening to an episode on the bus and at one segment in the show a story about her boyfriend's favourite director is told. Jessica wants to recommend the story to her boyfriend, who doesn't usually listen to podcasts. She presses the share icon below the episode. In the text box that appears next, she writes "you will like this" and presses the submit button. The podcast manager displays a green mark indicating that her sharing was successful.

- Task 2: Secondary user

Berit Hansen is an older woman but is still doing well and likes to be active when the weather is nice. She likes to go for long walks and play card games with her friends. She is very delicate with her hair, so she does not like to go out when it rains and storms. She lives a bit away from the nearest supermarket and suddenly it starts to rain. Berit has an iPad and has middle experience with using apps and other technologies. She has already tried asking for help with grocery shopping, and it saved her information automatically for later use. She hits the order button and now she waits patiently for her groceries to arrive.

3. Validating the tasks

- You need to find one or more target users.
- Especially important if the design team didn't interview people in the task identification.
- Revise task examples or discard them.

Personas

- Capture a set of user characteristics (user profile)
- Bring users to life
 - Name, stock photo, characteristics, goals, and personal background
 - Relevant to product under development
- Synthesized from real people based on user research
- Typical, not idealized



Julia's Cafe / Ideal User Persona

Janelle Robinson

Age: 26
Status: Undergraduate

Customer Profile: A busy PhD Student who needs a quiet place to study and read without distractions. She spends a lot of time on campus, refuels often and is a major coffee lover. She is the ideal customer for Julia's Cafe. She wants to receive quick and professional service; order online from her smartphone to avoid lineups, and not deal with over-conversational staff members.

Motivations • High quality, strong coffee. • Quiet atmosphere for studying. • On or close to the college campus.	Frustrations • Hates waiting too long for her coffee. • Doesn't want to be bothered by the staff while studying. • Actually relies on strong coffee to help her concentrate later in the day.
Goals • She needs information on the type of coffee a cafe serves, in order to determine the quality. • She wants to find the menu and the daily specials as quickly and easily as possible. • She wants to order with her smartphone, avoid the usual line-up, and experience minimal distractions.	Favorite Coffee Beans • Ethiopian Yirgacheffe Coffee • Sumatra Mandheling Coffee • Ethiopia Charbanta Natural

Personas: Why?

- Helps with design
 - Empathy in the design/development team
 - Justify design decisions
 - Recruiting and testing
- Easy way to communicate to others
 - Software developers
 - Marketing
 - Managers
- Very common in industry

Minimal Personas

Stock Photo

Primary goals

Age

Motivations

Demographic info

Pain points

Maybe a quote

Behaviors, daily routine



Most should be related to the situation
Ok to have a few extra info for empathy/believability

Example Persona

BRANDI TYLER

MUNRO
AMERICAN

PROFILE	Narrow Feet
GENDER	Female
AGE	36
LOCATION	Los Angeles, CA
OCCUPATION	Receptionist; \$38k

MOTIVATIONS

Brandi gets very emotional about shopping for shoes in retail stores because she rarely can find a pair that fits her narrow feet. Recently, she's turned to online shopping to avoid the hassle of shopping in stores. Brandi found Munro after Googling "narrow width shoes" and reading other reviews online about the company.

GOALS

- Needs an SS (4A) width shoe
- Would like to purchase several pairs to fit occasion, style, and color
- Hoping to find that she doesn't have to sacrifice style or options when searching by fit

FRUSTRATIONS

- Not being able to filter available shoes by width
- Getting far fewer options when she applies width filter
- No other recommended shoes when she's looking at a pair she particularly likes



"It's SO difficult to buy shoes that fit my feet."

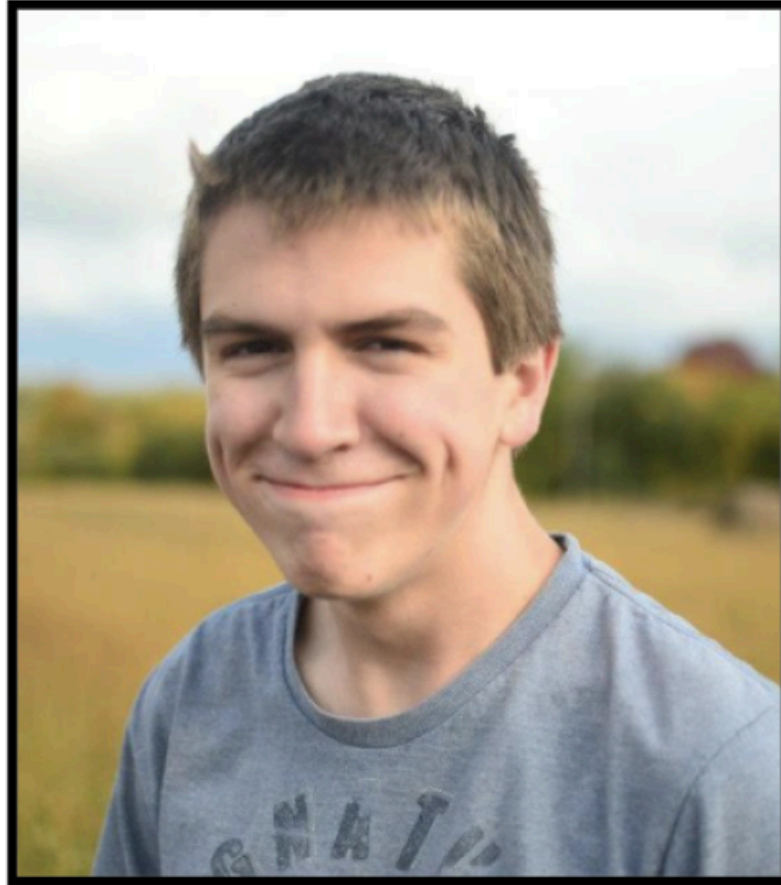
REAL MUNRO CUSTOMERS

"My whole life has been a choice between fit and style - when I was younger, I went for style & my feet killed me. As an adult, I tried for fit & the styles were for 95 year olds. This shoe is the 1st time I could get both."

"I wear a 4A and I have struggled my entire life finding shoes narrow enough for my feet and more so in recent years. I stumbled onto this Munro brand sandal and was shocked to find it comes in up to a 4A width and it actually fit and is like wearing a glove! I now have two pairs in different colors."

"Love these slides so much I went out and bought two more pairs. I have very narrow feet and they fit perfectly. They're very stylish and I get compliments whenever I wear them."

How good are these personas? Why?



Name: Asger

Age: 21

Gender: Male

Location: Amager

User story: Using the system to help elderly whenever he got extra time on his hands. Gives him a wonderful feeling helping elderly.

User Quotes: *"I like to help and the people I meet are always so nice and thankful. To be able to really make a difference, is something I will never grow tired of."*

Persona 1



Name: Eric Wang

Key Goals: Easily/quickly provide and access meaningful peer feedback, ensure anonymity, complete required course forms fast.

Needs: Web interface, easy navigation, filtering options, ability to comment, anonymity.

User Quotes: "I just want a platform where I can quickly rate teammates without worry of repercussions."

Age: 21

Persona 1: Romeen Haddad

Age and gender. 19, male.

Role. International student in the first year.

Background. Fluent in English but prefers icon first instructions. New to campus layout. Often checks maps at building entrances. Keeps screen brightness low to conserve battery.

Goals. Avoid getting lost and gain confidence when exploring a new campus. Recognize turns quickly in harsh sunlight. Enter buildings with clarity about next steps.

Design relevant needs. High contrast, large tap targets, icon first labeling. Brief voice cues such as "Turn right" and "Elevator ahead." Battery friendly decision point AR. Clear transition from outdoor to indoor.

Quote. "Give me simple icons and short cues, just the next turn."

Priority group. Must include.

4. Selecting User Types and Tasks

- Users

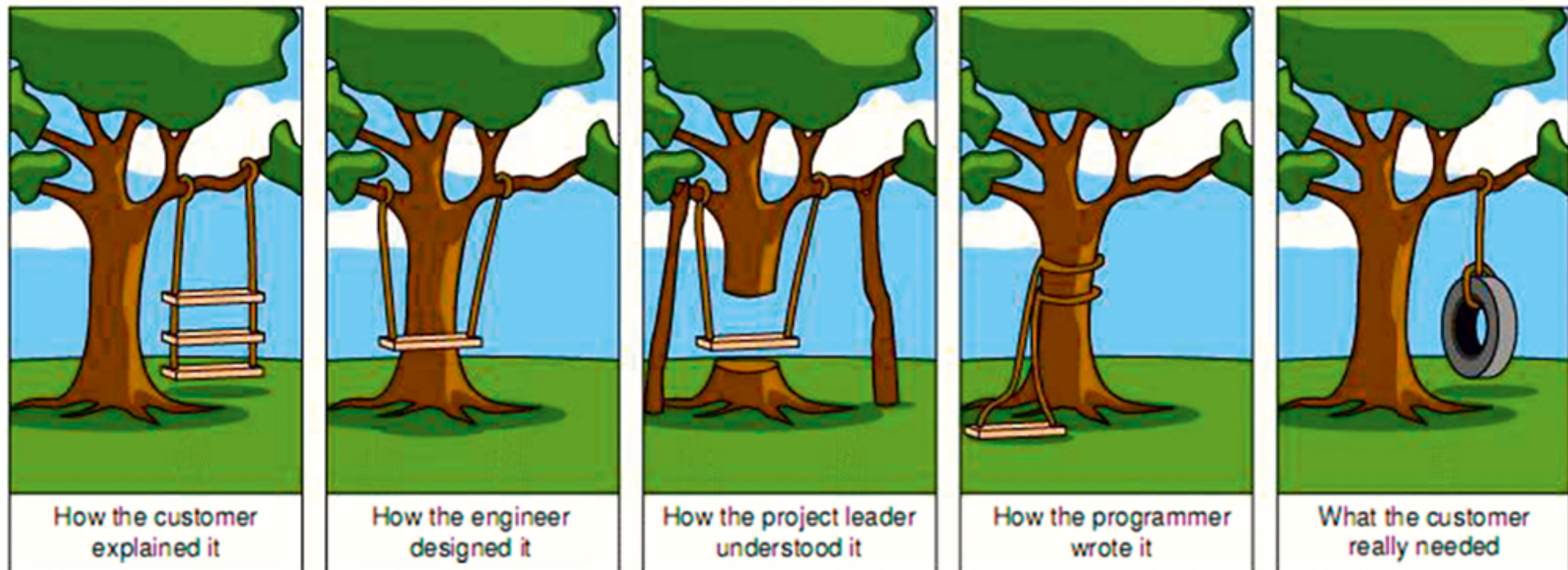
- **Absolutely must include:** main users
- **Should include if possible:** less important, atypical
- **Exclude:** rare, unimportant, different from main users

- Tasks

- **Absolutely must include:** frequent and important
- **Should include if possible:** pending time and budget, less frequent
- **Could include:** if you can add for “free”
- **Exclude:** unimportant or rare

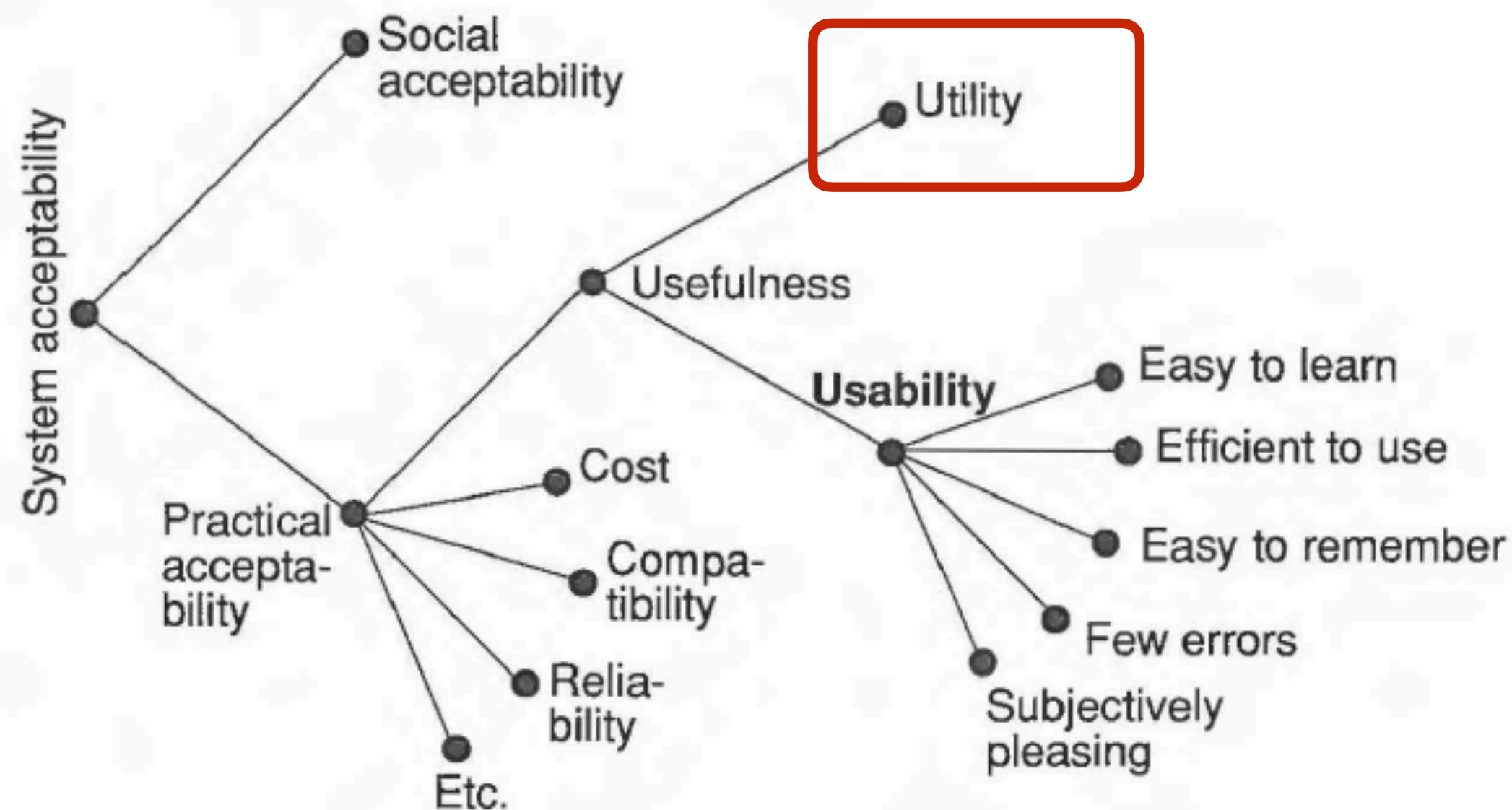
Discovering requirements: Why?

- Prevent miscommunications
- Help move from design to development
- Support collaboration among team members (e.g. between designers and software developers)



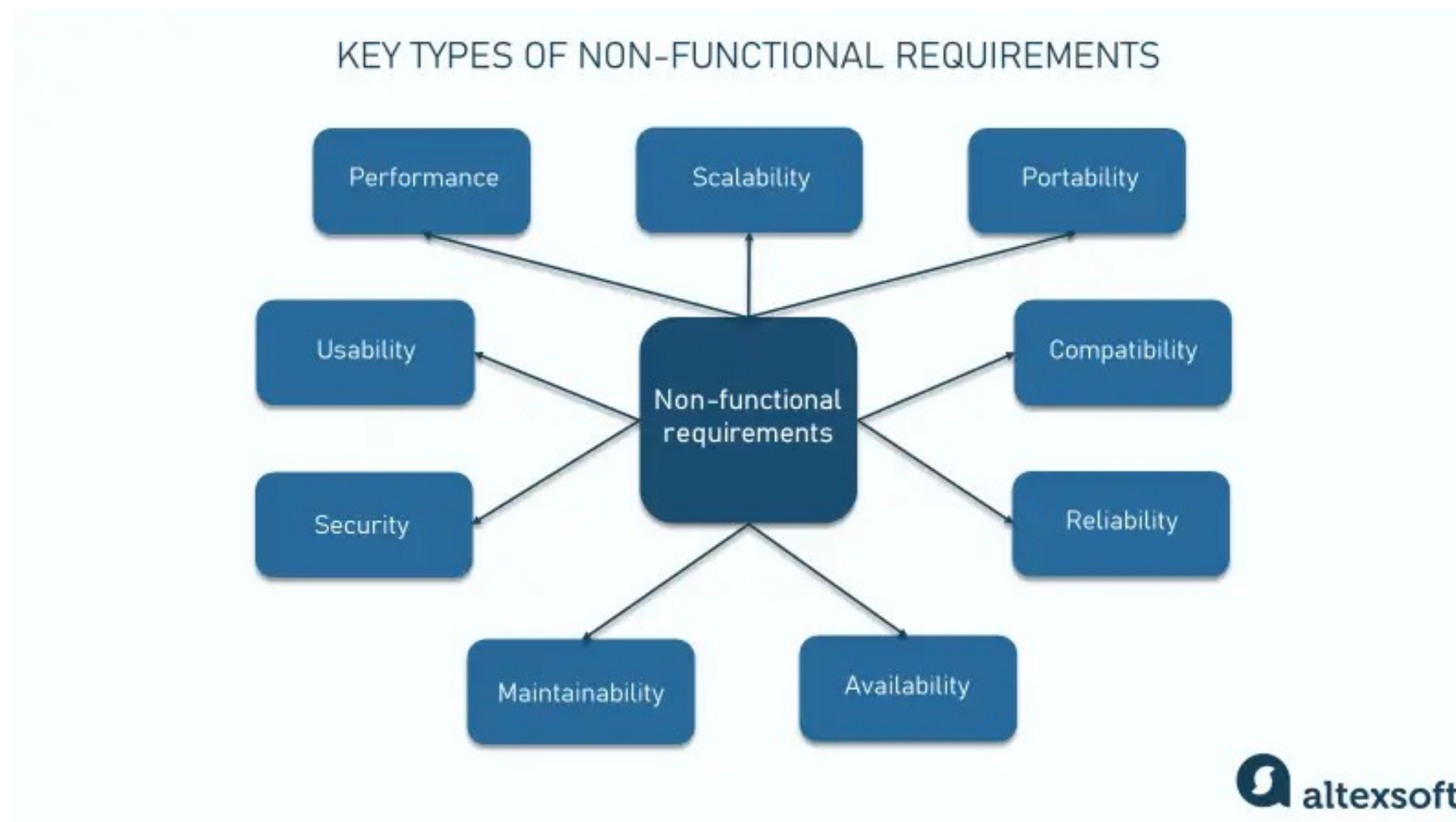
Defining requirements

- A requirement is **a functionality** or **a quality** that the product must have
- **Functional requirements** describe what the product will do
 - Historically, the main focus of requirements activities
- Jacob Nielsen: Utility is the question of whether the functionality of the system in principle can do what is needed.



Non-functional requirements

- Describe the **characteristics of the product**
- Covers various design aspects: visuals, usability, performance, security, cultural acceptability
- As important as functional requirements for the product's success



Example: Design an interactive system that helps a person find a roommate.

Discuss requirements:

- 2 functional
- 2 non-functional

Functional

The system must allow users to search for available rooms in a geographical area

The system must allow users to upload a description of their rooms

Non-functional

The system must be easy to learn.

The system must enhance sociability.

Bringing Requirements to Life

- A few approaches to capturing requirements in a human-centred way
 - Task descriptions
 - Personas
 - Several others (e.g., scenarios, customer journey maps, design fiction)